

Volkan Kilinc, Ph.D.

Scientist & AI Architect

Age: 32 years

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Professional Summary

Ph.D. Scientist and AI Systems Architect with 5+ years of R&D experience in DNA nanotechnology and biosensors. I have a proven ability to lead complex projects from **concept to peer-reviewed publication**, secure **competitive international funding**, and write patents. I now bridge deep science with technology by **architecting and building full-stack AI applications**. Most recently, I directed generative AI to deploy **Open AI Science**, a multi-agent platform for automated scientific peer-review. I am seeking to apply this unique blend of scientific expertise and AI product development to solve high-impact challenges.

Core Competencies & Technical Skills

❖ AI System & Product Management

- **AI System Architecture:** Designing and directing generative AI (Gemini) to build, test, and deploy complex, full-stack applications.
- **AI-Driven Development:** Managing the complete Software Development Lifecycle (SDLC) by steering AI for code generation, debugging, and visual design.
- **Technical Oversight:** Managing all product components, including database logic (SQL), real-time interactivity (WebSockets), and cloud deployment.
- **Prompt Engineering:** Developing sophisticated prompts to manage system logic, code generation, and technical outcomes.

❖ Research & Project Management

- **End-to-End Project Leadership:** Full lifecycle management from concept and staff coordination to peer-reviewed publication.
- **Funding Acquisition:** Proven success in grant writing (ERC, MSCA, SNSF) and securing fellowships (JSPS).
- **Technology Transfer:** Experienced in patent writing, IP strategy development, and market analysis.

❖ Scientific Domain Expertise

- DNA Nanotechnology & Self-Assembly
- Biosensors & Field-Effect Transistors (FETs)
- Surface Chemistry & Functionalization
- Drug Delivery & Molecular Encapsulation
- Soft Materials & Polymer Science

AI-based Projects Highlights:

Open AI Science - AI-Powered Scientific Peer Review Platform | Project Lead & Architect

Conceived, designed, and engineered "Open AI Science," (<https://www.openaiscience.com>) a full-stack web application that uses a multi-agent AI system to automate and enhance the scientific peer-review process. Distinct

"specialist" agents (e.g., Librarian, Analyst, Scorer) collaborate to deconstruct, analyze, and score manuscripts, providing objective and comprehensive feedback.

AI-Directed Software Engineering: Crypto Gaming App (RoadToMars) | Project Lead & Designer
Directed a generative AI (Gemini) to architect and build a full-stack, real-time Web3 application on the Solana blockchain (<https://www.roadtomars.app/>). Managed the complete DevOps lifecycle, overseeing the AI-assisted coding and independently deploying the application to cloud services.

Research Projects Highlights:

DNA-FET for High-Speed Data Retrieval | Lead Researcher & Inventor

Developed and fabricated a proof-of-concept DNA Field-Effect Transistor (FET), demonstrating a novel, non-destructive electronic method for data location in DNA data storage. Validated the sensor's high performance, leading to a first-author publication in *Advanced Sensor Research* and a Japanese patent application on the core technology. This foundational work established the basis for the proposed **AXON-DNA** system, a next-generation platform envisioned to integrate multi-sensor arrays with an AI-powered indexing engine for large-scale, random-access data retrieval.

DNA-pods: Biomimetic Nanocarriers for Drug Delivery | Lead Researcher & Inventor

Designed and synthesized "DNA-pods," a novel class of self-assembled DNA condensates, as a cost-effective platform for therapeutic delivery. Engineered a unique, thermally-triggered exfoliation mechanism inspired by viral uncoating to enable stimulus-responsive payload release. Validated the platform by demonstrating efficient encapsulation of doxorubicin and its preferential localization within the nucleus of fixed cancer cells.

Professional Experience

AI Development Consultant • Independent • Paris, France • Jun 2025 – Present

Formulating consulting offerings to help researchers and tech startups leverage AI for rapid prototyping, workflow automation, and innovative product development through vibe coding.

Postdoctoral Fellow (JSPS) • National Institute for Materials Science (NIMS) • Tsukuba, Japan • Apr 2022 – Apr 2025

Secured a highly competitive ¥3M JSPS Postdoctoral Fellowship to pioneer the use of DNA Field-Effect Transistors (DNA-FETs) for data storage applications. Managed the full research lifecycle, resulting in corresponding-author publications and patent filing. Invited Researcher from January to April 2025 to work on research initiatives in DNA-based 2D materials and drug delivery systems.

CTO & Co-founder • Gensor • Paris, France • Jun 2021 – Jun 2023

Co-founded a deep-tech startup to commercialize a proprietary virus testing platform based on gene field-effect transistor technology. Secured non-equity funding and mentorship by gaining acceptance into the competitive MassChallenge global accelerator program.

Project Manager, Technology Transfer • CNRS Innovation • Paris, France • Jan 2021 – Jan 2022

Performed due diligence on early-stage (TRL 1-2) research projects to assess commercial viability and guide funding strategy. Developed technology valorization roadmaps by conducting market analysis, end-user needs assessment, and intellectual property (IP) analysis.

Education

Ph.D., Nanoscience & Condensed Materials • Aix-Marseille Université • Marseille, France

M.Sc., Polymer Science (Magna Cum Laude) • Université Paris-Est Créteil • Créteil, France

B.Sc., Biology & Chemistry (Magna Cum Laude) • Université Paris-Est Créteil • Créteil, France

Conferences & Scientific Dissemination

Presented research at over 10 international conferences and workshops, including 5 oral presentations (one as an invited speaker) and a poster presentation to Nobel Laureates at the 15th HOPE Meeting in Kyoto (Japan).